

1. Testing Methodology

Evaluation is based on the 26-case Evaluation Dataset (Evaluation Dataset.xlsx), spanning technical surveys, comparative decisions, business decisions, vague/underspecified objectives, revision-triggering topics, and two deliberate configuration-failure cases. Each test case specifies an expected agent sequence, expected workflow path (including whether a revision loop should occur), and expected research task count, against which an actual run can be scored.

Two categories of metric are reported here: (a) metrics that can be computed by static inspection of the implementation (e.g. whether a hard revision cap exists), and (b) metrics that require executing real workflow runs against a live OpenRouter and Tavily configuration. This document is honest about which category each figure falls into — no numerical result below was fabricated. Where live execution has not been performed as part of producing this document, that is stated explicitly rather than presenting an invented number.

2. Evaluation Metrics

2.1 Task Planning Accuracy

Definition: the proportion of Supervisor-generated plans that contain 3-6 well-formed, non-overlapping research questions relevant to the stated objective.

Status: not yet measured experimentally. This requires running the Evaluation Dataset's objectives through a live Supervisor and manually or automatically scoring the resulting ResearchPlan against each row's 'Expected Research Tasks' column.

2.2 Agent Routing Accuracy

Definition: the proportion of runs whose actual node sequence (as recorded in execution_logs) matches the expected sequence in the Evaluation Dataset, including correct handling of the Critic-to-Analyst revision loop.

Status: not yet measured experimentally. Structurally, routing correctness for the non-revision path is guaranteed by the static graph edges (supervisor -> research -> analyst -> critic -> writer), so deviation would only be expected on the conditional edge out of the Critic node — this is the metric most worth prioritizing for a first measurement pass.

2.3 Workflow Completion Rate

Definition: the proportion of submitted runs that reach status='completed' rather than 'failed', excluding intentionally-invalid configuration test cases (EVAL-021, EVAL-022).

Status: not yet measured experimentally.

2.4 Evidence Coverage

Definition: the proportion of research questions in a plan that receive at least one non-empty evidence result from Tavily.

Status: not yet measured experimentally. Expected to be near 100% for mainstream topics and intentionally low for adversarial/fictional-topic cases like EVAL-011 and ADV-003, which test that low coverage is handled honestly rather than papered over.

2.5 Critic Detection Rate

Definition: the proportion of deliberately weakened AnalysisResult inputs (e.g. an insight with no supporting citation) that the Critic correctly flags as REVISION_REQUIRED.

Status: not yet measured experimentally. This metric depends on constructing synthetic 'flawed analysis' fixtures and observing the Critic's verdict, similar to the approach already used in the automated test suite's patched_llm fixture.

2.6 Handoff Success Rate

Definition: the proportion of the 7 documented handoffs (see Handoff Schemas.docx) that pass their schema validation on the first attempt, per run.

Status: not yet measured experimentally at the dataset level. Schema validation itself is enforced unconditionally by Pydantic for every handoff regardless of measurement — a run either produces a valid handoff or is marked failed; what is not yet measured is the empirical failure rate across the dataset.

2.7 Human Approval Compliance

Definition: N/A for this implementation. The current architecture has no human-approval checkpoint (see Known Limitations) — the Evaluation Dataset's 'Human Checkpoint' column is 'None' for every case, since revision decisions are made entirely by the automated Critic agent, not a human reviewer.

2.8 Average Workflow Time

Definition: mean wall-clock time from POST /api/workflows to status='completed', across successful runs.

Status: not measured. No timing instrumentation results are available at the time of writing this document; this should be measured directly from created_at/updated_at timestamps on the workflow_runs table across a batch of live runs.

2.9 Average Agent Calls

Definition: mean number of LLM calls (Supervisor + Analyst + Critic + Writer, counting each revision cycle) and Tavily calls (one per research question) per run.

Status: not measured. Can be computed directly by counting execution_logs rows per agent per run once a batch of live runs exists.

2.10 Approximate Cost

Definition: estimated OpenRouter + Tavily spend per completed run, based on the model and plan configured in OPENROUTER_MODEL and the number of calls in 2.9.

Status: not measured — depends on the specific model selected via OPENROUTER_MODEL (cost varies substantially by model) and on Tavily's pricing tier, neither of which is fixed in the codebase.

Should be computed after a batch of live runs using the OpenRouter dashboard's reported token usage rather than estimated from first principles.

3. Observations

- The system's error-handling design (typed exceptions, no mock/fallback data, a single outer exception boundary) makes failed runs easy to diagnose after the fact, which should make the experimental metrics above straightforward to collect once live runs are executed.
- The revision cap (`max_revisions`, default 2) is the one mechanism in this evaluation set that is guaranteed correct by code inspection alone rather than by observed LLM behavior — see ADV-007 in `Adversarial Tests.xlsx`.
- Several dataset cases (EVAL-004, EVAL-006, EVAL-024) specifically probe whether the Analyst's `comparison_table` is populated only when genuinely appropriate, which was called out in the source documentation as a previously-fixed defect ('forced fake comparison table').

4. Strengths

- Clear separation of concerns between agents, each with a narrow, independently testable responsibility.
- Fail-fast configuration checking prevents partially-run workflows from silently producing degraded output when a required API key is missing.
- Evidence is persisted immediately per-result, so partial progress is never lost to a later-stage failure — this materially simplifies debugging failed runs.
- A code-enforced (not LLM-enforced) termination guarantee on the revision loop removes an entire class of runaway-cost failure mode.
- The References section being deterministically generated from real database rows removes source hallucination as a risk in that specific part of the report, regardless of how the underlying LLM behaves.

5. Weaknesses

- No experimental data has yet been collected against the Evaluation Dataset — every quantitative metric above is currently a definition and a measurement plan, not a result.
- No clarification checkpoint exists, so vague objectives (EVAL-005) are handled entirely by the Supervisor's own judgment with no opportunity for the user to refine scope mid-run.
- Citation validation (`validate_citations()`) exists in code but is not wired into the automatic pipeline, so citation correctness within the report body (as opposed to the References list) is not currently enforced end-to-end.
- No rate limiting or per-user cost ceiling exists at the API layer.
- The tasks table is unpopulated, so there is no per-question progress visibility beyond what `execution_logs` happens to capture.

6. Conclusion

The Intellisys implementation has a sound architectural basis for the metrics defined in this document — most notably, several safety properties (the revision cap, evidence persistence, fail-fast configuration) are guaranteed by code structure rather than by observed model behavior, which is a stronger form of correctness than metrics that depend entirely on LLM output quality. However, this evaluation report should be read as a methodology and instrumentation plan at this stage: the numerical results for task planning accuracy, routing accuracy, completion rate, evidence coverage, critic detection rate, handoff success rate, timing, agent-call counts, and cost are all currently unmeasured and are flagged as such rather than estimated. The recommended next step is to execute the 26-case Evaluation Dataset against a live configuration and populate the Actual Result and Pass/Fail columns, from which every metric in Section 2 can be computed directly.